Taylor Cone Experimental Validation: Detailed Apparatus and Procedure

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1 Introduction

This document provides complete experimental apparatus specifications, detailed safety protocols, and step-by-step procedures for Taylor cone onset validation experiments. It is designed as a standalone reference for experimental implementation of the theoretical and computational work described in the main ISEF paper.

1.1 Document Purpose

- Complete equipment list with budget estimates and suppliers
- Detailed safety protocols addressing all hazards
- Step-by-step setup and operational procedures
- Troubleshooting guide and emergency procedures
- Required for institutional safety review and approval

1.2 Scope

This procedure covers:

- High-voltage electrospray apparatus assembly
- Camera-based Taylor cone imaging
- Voltage ramp experiments for onset detection
- Image processing and data analysis workflow

2 Complete Apparatus List with Budget

2.1 High Voltage Power Supply

Primary Option: Current-Limited Lab Supply

- **Item:** Glassman High Voltage PS/EH Series (0–10 kV, 1–2 mA)
- **Specifications:**
  - Output: 0–10,000 V DC
  - Current limit: 1–2 mA (critical for safety)
  - Regulation: $\pm 0.01\%$
  - Digital display and remote control
- **Cost:** \$1,500–2,500 USD
- **Supplier:** Glassman High Voltage, Newark Electronics, Mouser
- **Safety features:** Hardware current limiting, emergency shutdown, discharge circuit

Alternative Option: More Affordable

- **Item:** Spellman SL Series (0–5 kV, 1 mA)
- **Cost:** \$800–1,200 USD
- **Supplier:** Spellman HV Electronics, eBay (used units)

Budget Option: Regulated Flyback Module

- **Item:** XP Power CA/CB Series with external current limiter

- **Cost:** $200–400 USD
- **Supplier:** DigiKey, Mouser
- **Note:** Requires additional enclosure and safety circuitry

Recommended: Glassman or Spellman for lab safety compliance.

2.2 Syringe Pump

Primary Option: Precision Syringe Pump

- **Item:** Harvard Apparatus PHD 2000 or PHD ULTRA
- **Specifications:**
  - Flow rate range: 0.001 μL/h to 220 mL/h
  - Accuracy: $\pm$0.5%
  - Compatible with 0.5–60 mL syringes
- **Cost:** $2,000–3,500 USD
- **Supplier:** Harvard Apparatus, VWR

Alternative Option: More Affordable

- **Item:** New Era NE-300 Single Syringe Pump
- **Specifications:**
  - Flow rate: 0.73 μL/h to 2120 mL/h
  - Accuracy: $\pm$1%
- **Cost:** $800–1,200 USD
- **Supplier:** New Era Pump Systems (syringepump.com)

Budget Option: DIY Stepper Motor Pump

- **Item:** Stepper motor + lead screw + Arduino control
- **Cost:** $150–300 USD
- **Supplier:** Parts from Amazon, Adafruit
- **Note:** Requires calibration and flow verification

Recommended: New Era NE-300 (good balance of cost and precision).

2.3 Camera and Imaging System

Primary Option: High-Speed CMOS Camera

- **Item:** FLIR Blackfly S USB3 (BFS-U3-32S4M-C)
- **Specifications:**
  - Resolution: 2048$\times$1536 pixels
  - Frame rate: 118 fps at full resolution
  - Sensor: Sony IMX252 (3.45 μm pixel)
  - Interface: USB 3.0
- **Cost:** $600–900 USD
- **Supplier:** FLIR, Edmund Optics

Alternative Option: Mid-Range USB Camera

- **Item:** ELP USB Camera (5MP, 2592×1944)
- **Specifications:**
  - Resolution: 5MP
  - Frame rate: 15–30 fps
  - Manual focus
- **Cost:** $80–150 USD
- **Supplier:** Amazon, AliExpress

Budget Option: Webcam with Macro Lens

- **Item:** Logitech C920 + macro lens adapter
- **Cost:** $100–150 USD
- **Supplier:** Amazon, Best Buy
- **Note:** Limited resolution but sufficient for proof-of-concept

Recommended: FLIR Blackfly S (research-grade) or ELP camera (budget).

2.4 Budget Summary

Table 1: Research-Grade Setup ($7,000–10,000 USD)

Component	Cost (USD)
Glassman HV Power Supply	$2,000
Harvard Apparatus Syringe Pump	$2,500
FLIR Blackfly Camera	$800
Telecentric Lens	$1,500
LED Backlight	$150
Optical Breadboard Setup	$500
Metal Enclosure + Safety	$500
Needles, Electrodes, Hardware	$300
Chemicals and Consumables	$200
PPE and Safety Equipment	$200
Calibration Tools	$150
Miscellaneous	$200
Total	**$9,000**

Table 2: Mid-Range Setup ($3,000–4,000 USD)

Component	Cost (USD)
Spellman HV Power Supply	$1,000
New Era NE-300 Syringe Pump	$1,000
ELP USB Camera	$120
C-Mount Macro Lens + Extensions	$80
Huion LED Light Pad	$50
Optical Breadboard Setup	$400
Metal Enclosure + Safety	$400
Needles, Electrodes, Hardware	$250
Chemicals and Consumables	$150
PPE and Safety Equipment	$150
Calibration Tools	$100
Miscellaneous	$150
Total	**$3,850**

Table 3: Budget Setup ($1,200–1,800 USD)

Component	Cost (USD)
XP Power HV Module + Limiter	$300
DIY Arduino Syringe Pump	$200
Logitech C920 + Macro Adapter	$130
DIY LED Panel	$35
DIY Aluminum Frame	$80
Acrylic Box + Wire Mesh	$180
Needles, Electrodes, Hardware	$150
Chemicals and Consumables	$120
PPE and Safety Equipment	$120
Digital Calipers	$60
Miscellaneous	$100
Total	**$1,475**

3 Safety Protocols and Risk Mitigation

3.1 Primary Hazards

1. **Electric shock** (potentially lethal)
2. **High voltage arc/spark** (fire risk)
3. **Chemical exposure** (ethanol, solvents)
4. **Sharp needle** (puncture injury)

3.2 High Voltage Safety Protocol

3.2.1 Before ANY Experiment

Required checks:

☐ All participants have read and signed safety acknowledgment

☐ Supervisor present and briefed on emergency procedures
☐ Emergency contact numbers posted on wall
☐ Fire extinguisher (Class C) within 3 meters
☐ First aid kit accessible
☐ Workspace clear of flammable materials
☐ All personnel wearing safety glasses
☐ Electrical insulating gloves available
☐ Door interlock tested and functional
☐ Emergency stop button tested
☐ All ground connections verified ($< 1\Omega$ to earth)
☐ Current limit verified at 1 mA
☐ Only ONE person operates HV controls
☐ All observers maintain 2 m distance from enclosure

3.2.2 During Experiment

Operating rules:

- Door MUST be closed before applying HV
- NO adjustments while HV is on
- If adjustment needed: TURN OFF HV → wait 60s → open door → adjust
- Only HV operator touches controls
- All observers stay behind marked line (2 m from enclosure)
- Monitor for unusual sounds (buzzing, crackling) → shut down immediately
- Monitor for unusual smells (ozone, burning) → shut down immediately

3.2.3 After Experiment

Shutdown procedure:

- Turn voltage knob to zero (fully CCW)
- Turn off HV power supply main switch
- Wait minimum 60 seconds (capacitor discharge time)
- Open door (safe to enter)
- Use insulated stick to short HV terminal to ground (extra precaution)
- Only then touch equipment
- Disconnect HV cable from needle
- Cover exposed HV terminals with insulating caps

3.2.4 Emergency: Someone is Being Shocked

DO NOT TOUCH THE PERSON

1. **Press emergency stop button** (breaks HV circuit)
2. **If E-stop doesn't work:** Unplug HV power supply
3. **Wait 5 seconds** (let capacitors discharge)

4. **Only then:** Check victim, provide first aid, call emergency services

Hong Kong Emergency: 999 (Fire, Ambulance, Police)

3.3 Chemical Safety Protocol

3.3.1 Ethanol Handling

- **Hazard:** Flammable, eye irritant
- **PPE required:** Safety glasses, nitrile gloves, lab coat
- **Ventilation:** Use in well-ventilated area or fume hood
- **Storage:** In flammable-safe cabinet, away from HV equipment
- **Spill cleanup:** Absorb with paper towels, dispose in hazardous waste
- **Disposal:** Follow local hazardous waste regulations (do NOT pour down drain)

3.3.2 First Aid

- **Eye contact:** Rinse immediately with water for 15 minutes, seek medical attention
- **Skin contact:** Wash with soap and water
- **Ingestion:** Do not induce vomiting, seek medical attention immediately
- **Inhalation:** Move to fresh air

3.4 Current Limit Verification

CRITICAL SAFETY STEP:

1. **Set current limit to 1 mA:**
   - Adjust current limit pot/knob to minimum
   - Use multimeter in current measurement mode if possible
2. **Why 1 mA limit is critical:**
   - Current through human body:
     - >1 mA: Barely perceptible
     - >10 mA: Painful, cannot let go
     - >100 mA: Potentially lethal (ventricular fibrillation)
   - 1 mA limit provides safety margin
   - Still sufficient for Taylor cone formation (nA–μA typical)

4 Detailed Setup Procedure

4.1 Phase 1: Pre-Assembly Preparation (1–2 days)

4.1.1 Workspace Preparation

Select appropriate workspace:

- Lab bench or sturdy table (minimum 1.2 m $\times$ 0.8 m)
- Near grounded electrical outlet
- Good lighting
- Fire extinguisher accessible (Class C for electrical fires)

- First aid kit nearby
- Clear of flammable materials

Verify electrical safety:
- Confirm outlet is properly grounded (use outlet tester)
- Check circuit breaker rating (minimum 10 A recommended)
- Test ground continuity (should read $< 1\Omega$ to earth)

4.2 Phase 2: Mechanical Assembly (2–3 hours)

4.2.1 Enclosure Setup

1. **Position enclosure on bench:**
   - Leave 30 cm clearance on all sides for access
   - Ensure door can open fully
   - Mark locations for cable entry
2. **Ground the enclosure:**
   - Attach ground wire to metal frame
   - Run to building ground or ground rod
   - Verify continuity: $< 1\Omega$ to earth ground
   - Label: "SAFETY GROUND - DO NOT REMOVE"
3. **Install interlock switch:**
   - Mount magnetic switch on door frame
   - Wire normally-open contact to HV supply enable line
   - Test: Opening door should cut HV output immediately
   - Label switch: "SAFETY INTERLOCK"
4. **Install emergency stop button:**
   - Mount red mushroom button outside enclosure
   - Wire in series with HV enable
   - Test functionality before proceeding
   - Label: "EMERGENCY STOP"

4.2.2 Optical Breadboard Assembly

1. **Mount breadboard inside enclosure:**
   - Use rubber feet or vibration dampers
   - Ensure level (use spirit level)
   - Leave space for HV supply underneath or outside
2. **Install optical posts:**
   - Post 1: Camera mount (rear, centered)
   - Post 2: Needle/syringe holder (front-center)
   - Post 3–4: Electrode mount (front, vertically adjustable)
   - Tighten securely but avoid over-torquing

4.3 Phase 3: Electrical Connections (1–2 hours)

CRITICAL: All electrical work must be done with HV power supply UNPLUGGED

4.3.1 HV Power Supply Placement

1. **Position HV supply:**
   - **Outside** enclosure if possible (easier access, better cooling)
   - On rubber mat (electrical insulation)
   - Away from liquids
   - Ventilation unobstructed
2. **Verify all controls are at minimum/off:**
   - Voltage knob: Minimum (fully CCW)
   - Current limit: Minimum
   - Power switch: OFF
   - Main power: UNPLUGGED

4.3.2 HV Output Connection

1. **Select appropriate HV cable:**
   - Use HV-rated silicone insulated wire (20–30 kV rating)
   - Length: Minimize excess (2–3 m maximum)
   - Color: RED for high voltage
   - Do NOT use standard hookup wire
2. **Connect HV output to needle:**
   - Route HV cable through enclosure grommet
   - Connect to needle holder (must be conductive)
   - Use HV-rated alligator clip or crimp lug
   - Ensure connection is mechanically secure
   - Wrap connection with HV-rated electrical tape
   - Keep cable away from grounded surfaces (> 5 cm clearance)
3. **Label HV cable:**
   - Attach warning labels every 30 cm
   - "HIGH VOLTAGE - DO NOT TOUCH"
   - Include maximum voltage rating

4.3.3 Ground Connections

1. **Ground the extractor electrode:**
   - Connect thick ground wire (12 AWG minimum)
   - Run to common ground point
   - Verify continuity to building ground
2. **Create single-point ground:**
   - All grounds should connect at ONE point
   - This prevents ground loops

- Use ground bus bar or large terminal block
- Final connection to building ground or ground rod

5 Step-by-Step Experimental Protocol

5.1 Pre-Experiment Preparation (30 minutes)

5.1.1 Day Before Experiment

1. **Check equipment:**
   - Verify all components are functional
   - Check HV cable insulation (no cracks or damage)
   - Test camera and image capture
   - Verify sufficient liquid supply
2. **Prepare test liquid:**
   - For ethanol + NaCl:
     - Measure 100 mL ethanol in graduated cylinder
     - Dissolve 0.01–0.1 g NaCl (adjust for desired conductivity)
     - Stir until fully dissolved
     - Store in sealed bottle, label clearly
     - Record: date, concentration, operator name
3. **Clean equipment:**
   - Wipe down electrode with ethanol (remove residue)
   - Inspect needle for contamination
   - Clean optical surfaces (lens, backlight diffuser)

5.2 Voltage Ramp Protocol (30–60 minutes)

1. **Initial state:**
   - Voltage: 0 V
   - Pump: OFF
   - Camera: Recording
   - Note starting timestamp
2. **Start liquid flow:**
   - Set pump to 0.5–2 μL/min
   - Observe meniscus forming at needle tip
   - Wait until steady hemispherical droplet appears ($\sim$1–2 min)
   - If droplet falls: reduce flow rate
   - Record stable flow rate in metadata
3. **Close enclosure and enable HV:**
   - Close door (verify interlock engages)
   - Turn on HV power supply main switch
   - Verify enable LED is lit
   - Voltage should still be at zero

4. **Begin voltage sweep:**
   - **Step size:** 100–200 V per step
   - **Dwell time:** 20–30 seconds per step
   - **Procedure for each step:**
     (a) Increase voltage by one step
     (b) Read voltage display, announce/record value
     (c) Observe meniscus through camera feed
     (d) Classify state:
       – "Hemispherical" - no deformation
       – "Ellipsoidal" - slight elongation
       – "Conical-unstable" - cone forms but oscillates
       – "Conical-stable" - steady cone, no spray
       – "Spray" - visible jet/droplet emission
     (e) Make verbal annotation (recorded by camera audio)
     (f) Wait for full dwell time before next step
5. **Identify onset condition:**
   - **Onset voltage (V_0):** First voltage where stable cone appears
   - Hold at this voltage for 60 seconds
   - Verify stability (no jetting, no collapse)
   - Record voltage and timestamp
6. **STOP immediately if:**
   - Sparking/arcing occurs → shut down, investigate
   - Unusual odor → shut down, ventilate
   - Current meter shows > 1 mA → shut down, check connections

5.3 Shutdown Procedure (10 minutes)

1. **Power down:**
   - Turn voltage to zero
   - Turn off HV power supply
   - Wait 60 seconds
   - Stop syringe pump
   - Stop video recording
2. **Safe equipment access:**
   - Open enclosure door
   - Short HV terminal to ground (insulated stick)
   - Disconnect HV cable from needle
   - Remove test liquid syringe
3. **Clean up:**
   - Wipe any liquid residue from electrode
   - Dispose of used needle in sharps container
   - Cap and label used liquid (if reusing)

- Store liquids in flammable cabinet

4. **Data backup:**
   - Copy video files to backup drive
   - Verify file integrity (can play back)
   - Fill in metadata template
   - Save metadata as JSON file with same filename as video

6 Troubleshooting Guide

6.1 Problem 1: No meniscus forms at needle tip

Possible causes:

- Flow rate too low
- Needle clogged
- Liquid wetting properties poor

Solutions:

- Increase flow rate gradually ($0.5 \rightarrow 1 \rightarrow 2$ μL/min)
- Check for blockage: disconnect needle, run pump, verify flow
- Replace needle if clogged
- Check liquid surface tension (should be > 20 mN/m for stable meniscus)

6.2 Problem 2: No cone formation even at high voltage

Possible causes:

- Liquid not conductive enough
- Electrode spacing too large
- Voltage too low
- Poor electrical connection

Solutions:

- Increase dopant concentration (add more NaCl)
- Measure liquid conductivity (should be > 10 μS/cm)
- Reduce electrode spacing to 10–15 mm
- Check HV connection to needle (verify continuity)
- Increase voltage gradually (literature onset: 2–5 kV typical)

6.3 Problem 3: Sparking/arcing between needle and electrode

Possible causes:

- Voltage too high for given spacing
- Sharp edges on electrode
- Contamination on surfaces

Solutions:

- **Immediately reduce voltage to zero**
- Increase electrode spacing

- Polish electrode edges (remove burrs)
- Clean surfaces with ethanol
- Check for conductive debris (liquid droplets on surfaces)

7 Emergency Procedures

7.1 Emergency 1: Electrical Shock

If someone is being shocked:

1. **DO NOT TOUCH THE PERSON**
2. Press emergency stop button or unplug HV supply
3. Wait 5 seconds (let capacitors discharge)
4. Check victim:
   - Conscious? → Reassure, check for burns, seek medical attention
   - Unconscious? → Call 999 immediately, begin CPR if trained
5. Do not move victim unless necessary for safety

7.2 Emergency 2: Fire

Small fire (< 30 cm):

1. Press emergency stop (cut power)
2. Use fire extinguisher (Class C)
3. Aim at base of fire, sweep side to side
4. If extinguished: ventilate area, document incident
5. If spreads: proceed to large fire procedure

Large fire or spreading rapidly:

1. Evacuate immediately (do not stop for belongings)
2. Close door to lab
3. Pull fire alarm
4. Call 999 from safe location
5. Meet at designated assembly point
6. Do NOT re-enter building

7.3 Emergency 3: Chemical Spill

Small spill (< 100 mL):

1. Alert others in area
2. Put on gloves and safety glasses
3. Absorb with paper towels or spill kit absorbent
4. Dispose in chemical waste container
5. Wipe area with water
6. Ventilate area

Spill on person:

- Remove contaminated clothing immediately
- Rinse affected area with water for 15 minutes
- Seek medical attention if irritation persists

8 Pre-Experiment Safety Checklist

Print and complete before EVERY experiment session:

Date: _____ **Operators:** _____
Supervisor: _____ **Start time:** _____

8.1 Equipment Checks

- ☐ HV power supply: voltage at zero, current limit at 1 mA
- ☐ Interlock switch: tested and functional (open door → HV disables)
- ☐ Emergency stop: tested and functional (press button → HV disables)
- ☐ Ground connections: verified $< 1\Omega$ to earth
- ☐ HV cable: inspected for damage (no cracks, no exposed conductor)
- ☐ Enclosure: door closes properly, no gaps
- ☐ Camera: recording function tested
- ☐ Backlight: functional, no flickering
- ☐ Syringe pump: smooth operation, no leaks
- ☐ Liquid supply: sufficient for planned experiments, properly labeled

8.2 Safety Equipment

- ☐ Fire extinguisher: present, charged, within 3 meters
- ☐ First aid kit: present and accessible
- ☐ Safety glasses: all participants wearing
- ☐ Electrical insulating gloves: available at workstation
- ☐ Nitrile gloves: available
- ☐ Lab coats: all participants wearing

8.3 Workspace

- ☐ Area cleared of flammable materials (> 1 m from enclosure)
- ☐ Emergency contact numbers posted on wall
- ☐ Adequate ventilation (windows open or fume hood on)
- ☐ Clear path to exit (no obstructions)
- ☐ Observers maintain 2 m distance (marked line on floor)

8.4 Personnel

- ☐ All participants have read safety protocols
- ☐ All participants have signed safety acknowledgment

☐ Roles assigned (HV operator: _____, recorder: _____)
☐ Everyone knows location of E-stop and fire extinguisher
☐ Everyone knows emergency phone number: 999

8.5 Final Checks

☐ Supervisor present and briefed
☐ Mobile phone present (for emergency calls)
☐ No food or drinks in work area
☐ No loose clothing or jewelry near moving parts

Supervisor signature: _____ **Approval to proceed:** YES / NO

9 Appendix: Quick Reference

9.1 Onset Voltage Estimation

Use this to predict expected onset voltage for your geometry:

$$V_0 \approx \sqrt{\frac{4\gamma L}{\varepsilon_0 \ln(4L/r_{\text{needle}})}} \tag{1}$$

Where:

- γ = surface tension (N/m)
- L = electrode spacing (m)
- $\varepsilon_0 = 8.854 \times 10^{-12}$ F/m
- r_{needle} = needle outer radius (m)

Example calculation (ethanol, 21-gauge needle, 20 mm spacing):

- $\gamma = 0.022$ N/m (ethanol)
- $L = 0.020$ m
- $r_{\text{needle}} = 0.0004$ m (21-gauge OD = 0.8 mm)
- $\ln(4 \times 0.020/0.0004) = \ln(200) \approx 5.3$

$$V_0 \approx \sqrt{\frac{4 \times 0.022 \times 0.020}{8.854 \times 10^{-12} \times 5.3}} \tag{2}$$

$$\approx \sqrt{\frac{0.00176}{4.69 \times 10^{-11}}} \tag{3}$$

$$\approx \sqrt{3.75 \times 10^{7}} \tag{4}$$

$$\approx 6,100 \text{ V} \tag{5}$$

Expect onset around 6 kV for this geometry.

9.2 Hong Kong Supplier Contacts

Fisher Scientific Hong Kong

- Phone: +852 2407 2600
- Website: fishersci.com.hk
- Products: Chemicals, lab equipment, consumables

VWR International (Asia Pacific)

- Phone: +852 2922 2082
- Website: hk.vwr.com
- Products: Lab equipment, chemicals

RS Components Hong Kong

- Phone: +852 2610 3888
- Website: hk.rs-online.com
- Products: Test equipment, tools, components

Sham Shui Po Electronics Market

- Location: Apliu Street (), Sham Shui Po
- MTR: Sham Shui Po Station Exit C2
- Hours: Daily 10:00–20:00
- Best for: Cables, connectors, tools, basic electronics

10 Document Version Control

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